

Jatinder Sahota

✉ sahotaj2@myumanitoba.ca |  [Jatinder Sahota](#) |  [jsahota11](#) |  [jsahota11.github.io](#)

EDUCATION

B.Sc. Computer Science – Mathematics Joint Honours (Co-op)

Aug. 2022 – Present

University of Manitoba

Winnipeg, MB

- Minor in Statistics
- GPA: 4.11/4.50

AWARDS AND SCHOLARSHIPS

- NSERC Undergraduate Student Research Award
- Dean's Honour List
- Dr. & Mrs. Murdoch MacKay English Award
- UM Queen Elizabeth II Entrance Scholarship - Tier 1
- William Wujek Scholarship

TECHNICAL SKILLS

Languages:

- Self-study: Python, HTML/CSS, LaTeX
- Course work: C/C++, Java, JavaScript, R
- Familiar with: BASH

Frameworks/Libraries:

- Self-study: Tailwind CSS, NetworkX
- Familiar with: Flask, Vis.js, Astro.js

Tools & Platforms:

- Self-study: Git/GitHub, GitHub Pages, Neovim, Adobe Photoshop
- Course work: LLDB
- Familiar with: GitHub Actions, Vercel, VS Code, JetBrains IDEs, Adobe Illustrator and Premiere, MS Office

PROJECTS

Graph Generator

Feb. 2025 – Present

- Engineered and deployed a full-stack web app that dynamically generates graphs from user input.
- Processed user-defined graph parameters on the backend in **Python**, using **Flask** and **NetworkX**.
- Generated graph structures in **JavaScript**, using **Vis.js** and turning backend JSON into node-edge lists constrained by user configuration selections.

Personal Website

Apr. 2025 – Present

- Built and deployed a responsive personal website in **HTML** to showcase projects, experience, and technical skills.
- Integrated **Tailwind CSS** for utility-first styling to phase out custom CSS styles.
- Implemented automated deployment to **GitHub Pages** using **GitHub Actions**, ensuring updates to the site are published seamlessly.

exFAT File Reader and Shell (Course Project)

May 2025

- Built a custom shell in **C** integrating GNU core utilities, input/output redirection and Unix-style pipes.
- Implemented an exFAT file system reader handling mounting, file descriptors, and an Open File Table.
- Parsed formatted volumes to utilize the Main Boot Region, FAT chains, and general directory entries.

Greed – Console Game (Course Project)

March 2025

- Implemented an interactive ASCII-based game in **Java**, applying object-oriented design through abstract classes, encapsulation, inheritance, and multiple interfaces.
- Developed a robust and modular game framework with clean, maintainable code separating game logic, UI, and player interactions, while enforcing error handling.
- Built a user-friendly console interface with clear menus, input validation and a dynamic game board that reflects the updating game state.

EXPERIENCE

Math Help Centre Tutor

Sep. 2025 – Present

Department of Mathematics, University of Manitoba

Winnipeg, MB

- Assisted **20+** students with topics ranging from introductory algebra to advanced calculus, providing clear explanations and problem-solving guidance.
- Adapted tutoring strategies in real-time to meet the needs of diverse learners in an unstructured environment.
- Supported academic success in students by fostering understanding, promoting independent thinking and building foundation in mathematical reasoning.

Undergraduate Student Research Award

May 2025 – Aug. 2025

Department of Computer Science, University of Manitoba

Winnipeg, MB

Supervised by Dr. Stephane Durocher, GADA Lab

- Designed and analyzed a new algorithm for the Online Traveling Salesman Problem on the circle, utilizing theoretical results and applying extensive testing.
- Conjectured and investigated the optimality of the algorithm, employing mathematical reasoning and substantial literature review to refine conclusions.
- Summarized findings in a final report, capturing insights and key results.

AI Model Trainer

Jun. 2024 – Jul. 2025

DataAnnotation

New York, USA (Remote)

- Assessed and debugged **250+** AI-generated code chunks across multiple languages, providing model-specific feedback with unit tests and ideal solutions.
- Analyzed **100+** mathematical model responses for correctness and clarity, writing professional evaluations outlining mistakes and flaws in the responses.
- Interpreted project documentation independently to understand criteria and context, applying critical analysis to **10+** diverse project types.

Calculus 2 Tutorial Advisor

Jan. 2025 – Apr. 2025

Department of Mathematics, University of Manitoba

Winnipeg, MB

- Guided **20+** students through advanced calculus concepts by working through and explaining worksheets during weekly tutorials.
- Graded **600+** quiz and exam questions, ensuring fairness and accuracy while maintaining academic integrity throughout the evaluations.
- Invigilated **2** exams and **5** quizzes while proactively preventing academic dishonesty during the assessments.

Copy & Print Associate

May 2023 – Aug. 2024

Staples Inc.

Winnipeg, MB

- Fulfilled **700+** print orders, managed self-serve stations, and maintained inventory in a high-volume retail setting.
- Coordinated with team members to handle large bulk orders under tight deadlines, ensuring print quality standards are consistently fulfilled.
- Balanced in-person and online customer service responsibilities, including passport photos, package services, and both scheduled and on-demand printing jobs.

VOLUNTEER EXPERIENCE

Lounge Supervisor

Mar. 2025 – Present

Computer Science Students' Association

Winnipeg, MB

- Maintained a welcoming and safe environment in the student lounge by mediating conflicts and enforcing respectful behaviour.
- Built community by greeting newcomers and helping students feel comfortable in the lounge.
- Reported lounge issues (cleanliness, dangerous equipment, etc) to the appropriate channels to ensure space remained functional and welcoming.